

$t$  $100 \text{ (nm)}$ **29****B**

$$E = \frac{p^2}{2m} = \frac{\left(\frac{h}{\lambda}\right)^2}{2m}$$

$$E \propto \frac{1}{\lambda^2}$$

$$\frac{4E}{E} \propto \left(\frac{\lambda}{\lambda_f}\right)^2 \Rightarrow \lambda_f = \frac{\lambda}{2}$$